

IMPROVING FEATURE EXTRACTION IN THE AUTOMATIC CLASSIFICATION OF SEISMIC EVENTS. APPLICATION TO COLIMA AND ARENAL VOLCANOES

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ABSTRACT

Monitoring of precursory seismicity in volcanoes is the most reliable and widely used technique in volcano monitoring. Since a visual inspection by human operators is a tedious task in a non-stop monitoring process, Hidden Markov Models have been previously proposed to automatically classify the different types of volcano-seismic events. Mel Frequency Cepstral Coefficients were successfully used as feature vector in this continuous classification system. In this paper seven novel features to be included in the MFCC feature vector are proposed. A very elementary GMM-based classifier has been implemented in order to assess the efficiency of the proposed parameters. Results using hundreds of events recorded from stations situated at Colima (Mexico) and Arenal (Costa Rica) volcanoes show that the proposed features improve the recognition accuracy and therefore they may be relevant in continuous volcano-seismic event automatic classification.

Index Terms— Volcano-seismic events, classification, Hidden Markov Models, MFCC, feature vector

1. INTRODUCTION

Seismic signals have proven to be an important tool for controlling the state of a volcano and the vicinity of a possible eruption. The rate of occurrence of seismicity in an active volcano is high, with the presence of hundreds of events per hour in days or weeks. Since a visual inspection by human operators is an arduous task in terms of time, Artificial Neural Networks [1, 2] and Hidden Markov Models (HMM) [3, 4] have been previously proposed to automatically classify the different types of volcano-seismic events.

The first step of the recognition process is signal processing feature extraction, which converts the volcano-seismic waveform to a parametric representation with less redundant information. Feature extraction consists of choosing those features which are most effective for preserving class separability. Benítez et al. (2007) proposed to use Mel Frequency

Cepstral Coefficients (MFCC) as feature vector. They developed a complete seismic-event classification and monitoring system based on the seismicity observed at the Deception Island Volcano (Antarctica). Their system was suitable to operate in real time and capable of discriminating between different types of events with an accuracy of about 90%. Additionally to MFCC, novel features representing both time and frequency behaviors of the different volcano-seismic events are proposed in this work. In order to assess the effectiveness of the proposed features, a very elementary Gaussian Mixture Model (GMM) based classifier [5] has been implemented.

The rest of the paper is organized as follows. Section 2 describes the seismic-signal data set. Section 3 presents the GMM-based classifier using both MFCC and proposed features. In Section 4 we analyze the improvement provided by the proposed features. Finally, Section 5 summarizes the main conclusions.

2. DATA SET

The volcano-seismic signals used in this work were recorded from stations situated at Colima (Mexico) and Arenal (Costa Rica) volcanoes. Nine types of event classes were manually labeled by expert technicians [1, 3, 6]: silence (SIL), explosions (EXP), long-period events (LP), regional earthquakes (REG), volcano-tectonics events (VT), harmonic tremors (T1), spasmodic tremors (T2), collapses (COL) and lahars (LAH). Unlabeled segments containing noise or unknown events were previously removed from continuous records.

Table 1 shows the types and number of events recorded at Colima and Arenal volcanoes, together with statistical information about their duration (standard deviation, mean, maximum and minimum values). The number of events varied from 24 (lahars) to 719 (long period events) and from 44 (regional earthquakes) to 1000 (tremor) in Colima and Arenal volcanoes, respectively. The parameter Std/Mean indicates the temporal variability of each class. We can observe that some classes have greater temporal variability (such as regional earthquakes in Arenal Volcano) than others (such as volcano-tectonic events in Colima Volcano)

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Colima Volcano						
Event	Number	Duration				
		Min(s)	Max(s)	Mean(s)	Std(s)	Std/Mean(%)
SIL	465	9.34	1788.90	212.63	278.87	131.15
EXP	211	16.50	874.70	103.98	74.07	71.24
LP	719	12.00	501.00	45.17	21.19	46.92
REG	111	22.90	360.30	143.31	39.77	27.75
T2	175	18.84	4263.90	370.33	511.42	138.10
COL	217	17.16	298.18	89.76	57.01	63.51
VT	229	18.90	134.84	41.18	13.44	32.64
LAH	24	1230.00	3536.54	2142.73	607.75	28.36
Arenal Volcano						
SIL	999	9.84	563.04	85.68	76.71	89.53
EXP	791	5.00	463.24	31.14	26.37	84.69
REG	44	22.44	823.70	150.92	175.17	116.07
T1	1000	3.36	596.28	82.08	87.38	106.47
T2	1000	8.92	599.40	136.73	116.83	85.45
VT	193	3.26	301.82	20.46	23.87	116.67

Table 1. Events recorded in Colima (upper part) and Arenal (lower part) volcanoes.

3. RECOGNITION SYSTEM

The GMM-based classifier implemented in this work divides firstly the signal in overlapping windows. For each window a feature vector ($\vec{x}$) composed of both MFCC and proposed features is then extracted. This vector (corresponding to a temporal window) is finally classified into one of the different classes of seismic events (C_i). Therefore:

$$\vec{x} \in C_i \Leftrightarrow P(C_i|\vec{x}) = \max_j P(C_j|\vec{x}) \quad (1)$$

where $P(C_i|\vec{x})$ is the probability of each class C_i given the input feature vector $\vec{x}$. Applying the Bayes rule:

$$P(C_i|\vec{x}) = \frac{p(\vec{x}|C_i)P(C_i)}{p(\vec{x})} \quad (2)$$

where $p(\vec{x}|C_i)$ is the probability density function of the class C_i evaluated at the input vector $\vec{x}$; $P(C_i)$ is the a-priori probability of the class C_i ; and $p(\vec{x})$ is the a-priori probability density of the input vector $\vec{x}$. Since $p(\vec{x})$ is independent of the class and considering $P(C_i)$ equal for all the classes, we can write:

$$\max_j P(C_j|\vec{x}) = \max_j \frac{p(\vec{x}|C_j)P(C_j)}{p(\vec{x})} = \max_j p(\vec{x}|C_j) \quad (3)$$

Searching for the maximum of $P(C_j|\vec{x})$ is therefore equivalent to looking for the maximum of $p(\vec{x}|C_j)$. In order to measure the efficiency of the proposed features only one Gaussian was assigned to each class. Thus, the probability density function of each class can be expressed using the following equation:

$$p(\vec{x}|C_j) = \frac{1}{(2\pi)^{N/2}(|\Sigma_{C_j}|)^{1/2}} \exp\left(-\frac{(\vec{x} - \vec{\mu}_{C_j})^t(\Sigma_{C_j}^{-1})(\vec{x} - \vec{\mu}_{C_j})}{2}\right) \quad (4)$$

where $\vec{\mu}_{C_j}$ is the mean of the Gaussian j ; Σ_{C_j} is the covariance matrix of the Gaussian j ; the argument of the exponential is a matrix product; $(\vec{x})^t$ represents the transposed of $\vec{x}$; $\Sigma_{C_j}^{-1}$ is the inverse of the matrix Σ_{C_j} ; and $|\Sigma_{C_j}|$ is the determinant of Σ_{C_j} .

Two GMM-based classifiers were performed in this work, one for Colima and one for Arenal Volcano, respectively. Half of the events of each class was used as training events. These events allow the probability density function of each class to be defined. The other half was used for testing the proposed features.

3.1. MFCC features

Mel Frequency Cepstral Coefficients have been successfully used in volcano-seismic events recognition systems [3]. Figure 1 shows a block diagram of the MFCC extraction process. The features are generated in a 4-s overlapping window with a 2-s window shift. A 512-point fast Fourier transform (FFT) is used to compute the magnitude spectrum, which serves as the input of an emulated filter bank that consists of 16 triangular weighting functions that were uniformly spaced between 0 and 25 Hz. The overlap between adjacent filters is 50%. The filter-bank analyzer gives a measurement of the energy of the signal in a given frequency band. Then, the logarithm of the output filter bank energies is calculated, resulting in a 16-parameter feature vector. Since the log filter bank energies are highly correlated, it is necessary to apply a de-correlation transformation. Thus, the discrete cosine transform (DCT) is used to de-correlate the features and reduce the number of components of the feature vector from 16 to 13 coefficients. Finally, the feature vector is augmented with linear regressions of the features (derivatives and accelerations). All together 39 features are used, resulting in a 39 dimensional feature space.

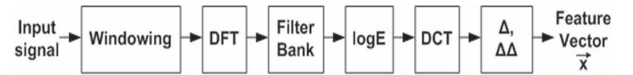


Fig. 1. MFCC feature extraction.

3.2. Proposed features

This work proposes seven novel features to be included in the feature vector. These features represent both time and frequency behavior of the different volcano-seismic events. Each temporal window of 4 seconds is therefore represented as a feature vector of 46 components. Figure 2 illustrates the proposed features:

- Impulsiveness of the signal onset. Normalized cumulative sum function of the absolute value of the signal may characterize the onset behavior of the event. The

	MFCC features								Nt	Nc	Acc(%)
	SIL	EXP	LP	REG	T2	COL	VT	LAH			
SIL	23627	0	307	329	201	0	9	39	24512	23627	96.39
EXP	57	3241	29	636	686	156	122	360	5287	3241	61.30
LP	683	133	4501	420	857	59	898	49	7600	4501	59.22
REG	72	350	114	2545	383	127	172	15	3778	2545	67.36
T2	791	2328	634	2761	7976	183	217	1048	15938	7976	50.04
COL	0	14	3	39	6	3650	38	997	4747	3650	76.89
VT	11	70	320	84	159	89	1434	2	2169	1434	66.11
LAH	907	12	131	84	50	615	84	10022	11905	10022	84.18
TOTAL									75936	56996	75.06

	MFCC features + proposed features								Nt	Nc	Acc(%)
	SIL	EXP	LP	REG	T2	COL	VT	LAH			
SIL	23755	0	325	116	292	1	4	19	24512	23755	96.91
EXP	83	3832	32	350	504	177	152	157	5287	3832	72.48
LP	419	125	5103	473	667	77	730	6	7600	5103	67.14
REG	8	432	130	2715	189	129	152	23	3778	2715	71.86
T2	711	791	590	1528	11208	228	81	801	15938	11208	70.32
COL	0	3	2	89	8	3754	575	316	4747	3754	79.08
VT	3	55	369	95	60	94	1493	0	2169	1493	68.83
LAH	886	5	131	55	59	727	89	9953	11905	9953	83.60
TOTAL									75936	61813	81.40

Table 2. Results for Colima Volcano.

50th-percentile (feature F1) is expected to be smaller in a slow onset than in a fast onset. Similarly, the difference between 70th and 30th-percentiles (feature F2) is expected to be greater in an event with a slow onset. Both features (F1 and F2) were calculated using a 6-s window.

- log-log frequency behavior. Events related to brittle fracture of solid rock (REG and VT events) present a log-log frequency spectrum characterized by a flat behavior up to a corner frequency (f_c), followed by a linear decay [7, 8]. The corner frequency is generally greater in VT events since the fracture zone is smaller than in REG events. This behavior can be fitted by a model that takes into account the corner frequency (feature F3) and the root mean square error between the spectrum and the model (feature F4). Feature F3 may then be used to distinguish between REG and VT events while feature F4 is expected to be useful to separate the REG and VT events from other classes. A 12-s window was used to compute the log-log frequency spectrum.
- Periodicity level. Normalized auto-correlation may be used to estimate the periodicity level of the different volcano-seismic events. The first maximum of the normalized auto-correlation function will be always equal to 1. The second maximum (feature F5) becomes progressively smaller as the signal becomes less periodic. This feature is then expected to be a periodicity level indicator of the seismic event. A 12-s window was used to calculate this feature.
- Presence of harmonics. Normalized cumulative sum function may be used to determine if an event presents a

dense power spectral density (PSD) or a PSD composed by delta functions (harmonics). The 50th-percentile (feature F6) and the difference between 70th and 30th-percentiles (feature F7) are expected to be smaller in a PSD composed by harmonics. A 10-s window was used to compute these two features.

4. EXPERIMENTAL RESULTS

The test events considered in this work were automatically classified using the classification system described above. Tables 2 and 3 show the confusion matrix using only the MFCC features (upper panel) and including the proposed features (lower panel) for Colima and Arenal volcanoes, respectively. The types of events, the total number of temporal windows corresponding to each event type (Nt), the number of windows correctly classified (Nc) and the accuracy (Acc) are indicated. Note that the automatic recognition system classifies each temporal window of 4 seconds in a volcano-seismic event.

We can observe that the accuracy is improved in all event classes by including the proposed features, except for classes LAH and VT in Colima and Arenal volcanoes, respectively, in which the accuracy is slightly reduced. Our results show that the MFCC features allow to classify the events with a mean accuracy (Acc) of 75.06% and 71.27% in Colima and Arenal volcanoes, respectively. Using all the features presented in this work the mean accuracy can be increased to 81.40% and to 84.87% in Colima and Arenal volcanoes, respectively.

	MFCC features						Nt	Nc	Acc(%)
	SIL	EXP	REG	T1	T2	VT			
SIL	15249	242	109	274	77	283	16234	15249	93.93
EXP	191	2848	683	1025	846	148	5741	2848	49.61
REG	489	197	440	225	162	71	1584	440	27.78
T1	4008	1352	417	2907	3303	67	12054	2907	24.12
T2	257	1329	1558	1964	27019	113	32240	27019	83.81
VT	62	46	201	38	7	524	878	524	59.68
TOTAL							68731	48987	71.27

	MFCC features + proposed features						Nt	Nc	Acc(%)
	SIL	EXP	REG	T1	T2	VT			
SIL	15619	85	60	135	92	243	16234	15619	96.21
EXP	118	3621	713	482	698	109	5741	3621	63.07
REG	368	293	559	145	136	83	1584	559	35.29
T1	1274	369	117	9518	758	18	12054	9518	78.96
T2	250	1758	831	864	28507	30	32240	28507	88.42
VT	28	68	242	17	13	510	878	510	58.09
TOTAL							68731	58334	84.87

Table 3. Results for Arenal Volcano

5. CONCLUSIONS

This work confirms that the MFCC represent the frequency behavior of the volcano-seismic events and therefore they are useful as feature vector in the automatic classification sys-

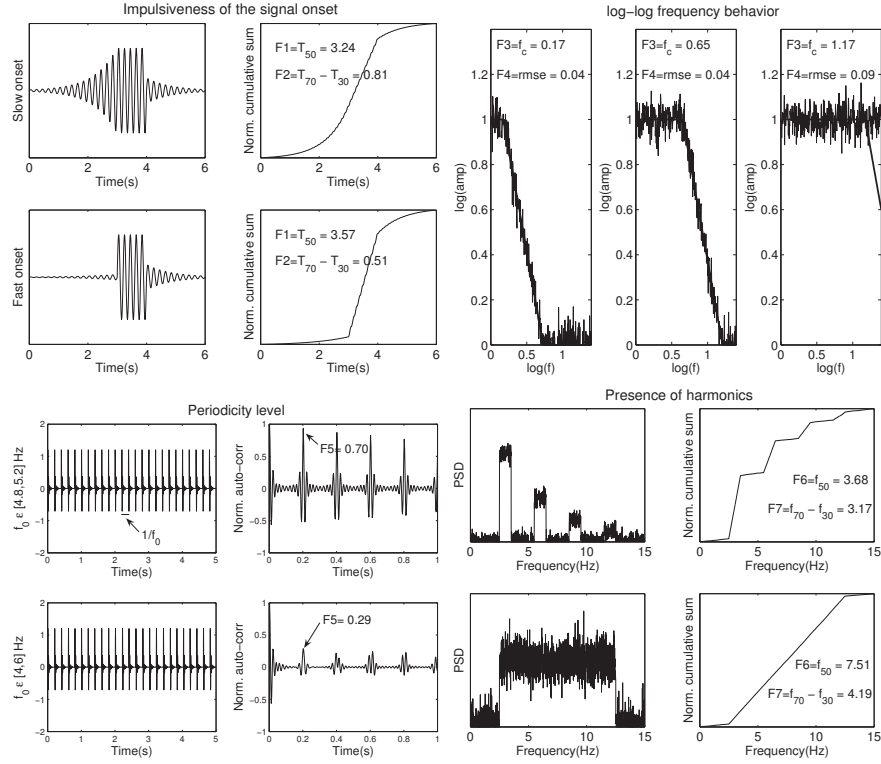


Fig. 2. Proposed features.

tems, which is in concordance with other studies. In addition to MFCC, seven novel features are proposed in this work. Our preliminary results suggest that the proposed features improve the recognition accuracy of an elementary GMM-based classifier and therefore they may be relevant in continuous seismic-event classification. Testing the proposed features in a continuous Hidden Markov Models -based recognition system is needed.

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